

Session 5: Crash course in data wrangling

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- 1 Introduction
- 2 Data wrangling
- 3 Any questions?
- 4 Working on projects

Introduction

Learning goals of today

- Short repetition from last week

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- Discussing “replication crisis” and potential solutions

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- Short repetition from last week
- Discussing “replication crisis” and potential solutions
- Crash course in data wrangling

Repetition: Useful commands

- `mutate()` [+ `case_when`]: Change data and create new variables

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Repetition: Useful commands

- `mutate()` [+ `case_when`]: Change data and create new variables
- `filter()`: Filtering data by specific values
- `select()`: Selecting variables

Repetition: Piping in tidyverse

```
# "Piped" code (as in the tidyverse)
iris_data <- iris |>
  filter(Species = "virginica") |>
  mutate(Petal.Length = Petal.Length + 1)
```

Reading - Reproducible science

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- What is a pre-analysis plan?

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Reading - Reproducible science

- What is p-hacking
- What is a pre-analysis plan?
- What is publication bias?
- How can bias creep into your research?

Reading - Suggestions they provide

- Improving methodological training

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Reading - Suggestions they provide

- Improving methodological training
- Encouraging collaboration and team science
- Promoting study pre-registration
- Improving the quality of reporting
- Promoting transparency and open science (data sharing)



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Data wrangling

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- Most of your time you will spend data wrangling
- Analysis itself will take less than 10% of your time
- Goal: To create clean, “tidy” data

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- Each variable is a column; each column is a variable.
- Each observation is a row; each row is an observation.
- Each value is a cell; each cell is a single value.
- For more information on tidy data:
<https://tidyr.tidyverse.org/articles/tidy-data.html>

group_by() - Applying manipulation by groups of data

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group_by() - Applying manipulation by groups of data

- Often you will need to manipulate data grouped by another variable
- Example: You have individuals, but you want to know how many people live in each city
- group_by() can be inserted before mutate operation in pipe
- group_by() manipulates data until ungroup() called



group_by() - Variable manipulation (code)

```
# Create sums by Species
iris_data <- iris |>
  group_by(Species) |>
  mutate(sum_Sepal.Length = sum(Sepal.Length)) |>
  ungroup() # always (!) remove grouping
```



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- Alternative: add .by = at the end of tidyverse command
- Only operates on this one function
- I prefer .by = over group_by()

.by = Alternative to group_by() (code)

```
# Create sums by Species
iris_data <- iris |>
  mutate(sum_Sepal.Length = sum(Sepal.Length), .by = Species)
```



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- load your data sets

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- load your data sets
- use `mutate()` together with `group_by()` or `.by` to create two new variables

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- join() commands necessary for combining data sets
- requires key variable to join on in both data sets!
- left_join(a,b) join b onto a, most common version
- others are right_join(), inner_join(), semi_join(), anti_join(), full_join()

join() - Combining different data sets - join_by()

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- key variable(s) need to be uniquely identify
- add join_by() after comma behind left_join()
- join_by(x == y) will join data set a and b using variable x from data set a and y from data set b
- Multiple join_by(x == y, h == j) possible

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- different kinds of joins possible
- one-to-one, many-to-one, one-to-many, many-to-many (the latter hopefully never!)
- relationship = “one-to-one” after comma behind `left_join()` to prevent errors

join() - Combining different data sets (code)

```
# Check whether keys identify uniquely (duplicated(sum) has
sum(duplicated(band_members_data$name))
sum(duplicated(band_members$name))

# Create sums by Species in band_members_data
band_members_data <- band_members |>
  left_join(band_instruments, join_by("name" == "name"), re
```



Test: join() - Combining different data sets (code)

- Do you have multiple data sets you would want to use?

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- use `left_join()` together with `join_by()` to create two new variables

arrange() - sort (multiple) variables

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- arrange() sorts variables from highest to lowest
- can be inversed from lowest (top) to highest using desc()
- can take multiple variables

arrange() - sort (multiple) variables (code)

```
# Sort entries by Petal.Width and Sepal.Width
iris_data <- iris |>
  arrange(Petal.Width, Sepal.Width)

View(iris_data)
```

Test: arrange() - sort (multiple) variables

- Try to sort your data set using first one, then two, then three variables

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- Often data will not come in tidy form
- there are two pivot options
- `pivot_longer()` - moves data from columns into fewer variables with more rows
- `pivot_wider()` - moves data from rows into more variables with fewer rows

pivot_longer() - example - before...

Name	Week1	Week2	Week3	Week4
Caroline	2	3	6	3
Jack	5	6	7	5
Ariana	3	4	3	7

`pivot_longer()` - example - ...and after

Name	Week	Amount
Caroline	1	2
Caroline	2	3
Caroline	3	6
Caroline	4	3
Jack	1	5
Jack	2	6
Jack	3	7
Jack	4	5
Ariana	1	3
Ariana	2	4
Ariana	3	3
Ariana	4	7

`pivot_wider()` - example - before...

Name	Week	Amount
Caroline	1	2
Caroline	2	3
Caroline	3	6
Caroline	4	3
Jack	1	5
Jack	2	6
Jack	3	7
Jack	4	5
Ariana	1	3
Ariana	2	4
Ariana	3	3
Ariana	4	7

`pivot_wider()` - example - ...and after

Week	Caroline	Jack	Ariana
1	2	5	3
2	3	6	4
3	6	7	3
4	3	5	7

pivot() - “switch columns with rows”

- pivot can be quite complicated

`pivot()` - “switch columns with rows”

- `pivot` can be quite complicated
- for more information, see `?pivot_longer()` or `?pivot_wider`

Any questions?

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